

ACHMAD RIVALDI ZULFAH

MACHINE LEARNING DEVELOPER | AI RESEARCH SPECIALIST

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Final-year Information Technology student (GPA 3.93/4.00) specializing in natural language processing and large language models. Experienced managing end-to-end ML pipelines — from data acquisition (web scraping) and text preprocessing through comparative experiments with IndoBERT and FLAN-T5 — and in shipping automation systems that run unattended in production, not just in a notebook.

KEY SKILLS

- **Machine Learning:** Supervised Learning (Classification & Regression), Model Evaluation (Accuracy, F1-Score, Precision-Recall).
- **NLP & LLM:** IndoBERT, Transformers (HuggingFace), Prompt Engineering (Zero-Shot, Few-Shot, CoT), Text Preprocessing.
- **Programming & Tools:** Python (Pandas, NumPy, Scikit-Learn), PyTorch, SQL, Git/GitHub, Google Colab.
- **Data & Automation Engineering:** Web Scraping (Playwright, Selenium, BeautifulSoup), local LLM inference (Ollama), workflow notifications (Telegram Bot API), Exploratory Data Analysis, Feature Engineering.

KEY ACHIEVEMENTS

- **State-of-the-Art Model Optimization.** Fine-tuned IndoBERT for Indonesian-language sentiment analysis, reaching 89.41% accuracy (F1-Macro 0.6070) — outperforming zero/few-shot and Chain-of-Thought prompting on FLAN-T5.
- **Production Automation Pipeline.** Built and deployed an unattended pipeline that scrapes job listings, scores fit with a locally-run LLM, and delivers real-time notifications — actively running, not a one-off script.
- **Comprehensive Data Pipeline.** Built an automated scraper and data-cleaning pipeline processing thousands of reviews from Indonesia's largest ride-hailing platforms (Gojek, Grab, Maxim).

PROJECT EXPERIENCE

Job Application Automation Pipeline (Independent Project)	2025 - Now
• Built an end-to-end pipeline: Playwright scrapes new Jobstreet listings, a locally-run LLM (Ollama) scores each for fit, and matches trigger a Telegram Bot API notification — keeping a human in the loop rather than auto-applying blind. • Designed for unattended, continuous operation; the system has been actively running in production against live listings. • Reports total listings scanned and applications sent as the pipeline's proven output; deliberately excludes win rate / interview outcomes from performance claims, since those depend on employer-side decisions outside the system's control. • Repo: github.com/Shoryuuuuuuu/job-automation	
AI Researcher & ML Developer (Undergraduate Thesis)	March 2026 - Now
• Conducted a comparative study evaluating fine-tuning vs. prompt engineering on LLMs for sentiment classification of user reviews. • Fine-tuned IndoBERT to handle slang and non-standard language in Indonesian ride-hailing reviews. • Tested Chain-of-Thought and few-shot prompting on FLAN-T5 to probe the limits of generative models on local dialects. • Result: IndoBERT (fine-tuned) was the strongest approach — Accuracy 0.8941, F1-Macro 0.6070. Identified and documented two root causes behind weaker few-shot/CoT results: an English-language few-shot/Indonesian-input mismatch, and CoT outputs missing the parser's expected "Final Answer" marker within the token limit.	

EDUCATION

Bina Sarana Informatika University (UBSI)	2022 - Now
Bachelor of Science in Information Technology GPA: 3.93 / 4.00	
• Focus: Natural Language Processing & Large Language Models.	
SMK Negeri 46 Jakarta	2018 - 2021
Software Engineering	

ADDITIONAL INFORMATION

Certifications: Coding Camp powered by DBS Foundation (Machine Learning Track).
Supporting Skills: Model Deployment, REST API Integration, Docker (Basic). | **Languages:** Indonesian (Native), English (Professional).